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Innovative Completion Methods to Overcome Challenging Multi-Stage Stimulation Completion Runs – North Sea Case Study

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Abstract

In the late 80's and early 90's, extensive hydraulic fracturing campaigns were executed to fracture many vertical wells in southern North Sea tight gas fields using the industries newest technologies at the time. In recent years, to boost production across two fields, campaigns involving modern horizontal, multi-stage stimulation completions were planned and executed to exploit the reserve pockets missed from historically only completing vertical wells.

Three horizontal wells were planned in two fields in the Southern North Sea sector; two sidetracks and one workover for a well that was drilled to fracture in 2015, though the hydraulic fractures were never executed. Considerable efforts were put into detailed reviews of previously executed treatments in vertical wells to develop fracturing geomechanics, and the optimum fracture designs. Treatments were designed to utilize the same proppant and similar fracturing fluid recipes where applicable to optimize resource planning. All completions were multi-stage ball actuated completions to minimize interventions between treatments, with key learnings from successful operations in other high profile North Sea assets applied to increase operations efficiency.

The wells were completed and campaigns executed in 2023. One well was successfully drilled and completed as planned, with all frac sleeves actuated and injections carried out in every sleeve. The multi-stage completion did not reach TD in the workover well, and real time decisions on alternative fracture sleeve locations were necessary, the result being the placement of only a single hydraulic fracture. The final new well involved an extensive fishing operation with drill pipe stuck in hole and required an inventive solution to save the well. A chemical cut in an optimum location for hydraulic fracture initiation was performed to avoid high-cost side-track operations. Fracturing strategies were quickly adapted to execute a single hydraulic fracture through the fish with an engineered over-displacement to remove the need to run coiled tubing in a high-risk non-conventional completion. All three wells produced remarkably well; the multi-stage treatment came online as the best producer in the field, and the two wells with the adapted completions met close to the well P90 production estimates with a single hydraulic fracture in each. This paper discusses the planning and execution of a fully utilized multi-stage completion, and the unique, real-time adaptations

made in the high stakes offshore environment, as the operations evolved for two other wells to still achieve significant production uplift in these fields.

Background

In the late 1980's and early 1990's, operators in the Southern sector of UKCS developed tight gas fields through hydraulic fracturing. The wells in the fields discussed in this paper were either vertical, or S-shaped and close to vertical through the pay, the Lower Leman sandstone. Completions configurations included dedicated packerless strings for fracturing, and treatment designs tended to be of large volume with aggressive tip-screen out methods, resulting in high screen-out rates (Bocaneala et al, 2015). Seawater based fracturing fluids were used, due to the typical job designs using large volumes of fluid for calibration injection and pad volumes, and the finite capacity of offshore stimulation vessels. The production uplift due to hydraulic fracturing ensured the fields were economically viable.

Since these campaigns ended several decades ago, no wells have been drilled for hydraulic fracturing in the Lower Leman sandstone targets of the fields in question, and as such, overall field production declined. Consequently, multiple campaigns were planned to boost the production of the two fields, Hydraulic fracturing strategies were updated with the modern multi-stage completion techniques, and the three wells planned would be the first hydraulically fractured horizontal wells in these fields.

Candidate Wells

Field A – Well A

Field A in the Southern North Sea sector is a tight gas field with an extensive natural fracture network. Field A consists of two compartments; a main compartment with several existing vertical wells, most of which were previously hydraulically fractured, and a southern compartment containing two wells which were both hydraulically fractured. There are three high quality reservoir units within the interval.

The existing well, Well A, at the time of planning the campaign, was an existing horizontal well in the southern compartment, which was drilled in 2015. Typical properties are shown in Table 1. The planned operations when the well was originally drilled included drilling a 3,000 ft horizontal section, cleaning and acidizing the open hole section, and installing a multi-stage completion for hydraulic fracturing. However, some issues were encountered during drilling, including tool failure, causing costly delays. Furthermore, there was a risk of borehole damage due to additional trips in hole, which may have caused issues when setting the packers in the completion system, therefore the programme only went as far as the acidizing of the open hole section. Initial production rates were promising after acidizing, however, rapid production decline was observed to around 25% of the initial rates. As such, Well A was considered as a candidate for re-completion for hydraulic fracturing.

Table 1—Typical Field A Southern Compartment Offset Well Reservoir Properties

Reservoir Parameter	Value
Lithology	Lower Leman Sandstone
Reservoir Temperature, °F	188
Reservoir Pressure, psi	4700
Porosity, %	5 – 20
Est. Permeability, mD	0.1
Est. Fracture Gradient, psi/ft	0.80 – 0.84

Field B – Wells B1 and B2

Field B in the Southern North Sea sector is a tight gas field, with an extensive history of vertical or S-shaped hydraulically fractured wells. Horizontal wells B1 and B2 were planned to exploit the northern most region, the tightest area of the field, with few wells. Wells B1 and B2 would be side-tracks of existing wells (on the same production platform), both of which provided sufficient offset well data, shown in Table 2. Planned horizontal sections for wells B1 and B2 were approximately 3,200 and 4,600 ft long respectively, to land in the best reservoir quality unit central to the interval.

Table 2—Typical Field B Northern Area Offset Well Properties

Reservoir Parameter	Value
Lithology	Lower Leman Sandstone
Reservoir Temperature, °F	213
Reservoir Pressure, psi	4500
Porosity, %	12
Est. Permeability, mD	0.05 – 0.15
Est. Fracture Gradient, psi/ft	0.57 – 0.61

Design

Completion selection

Planning commenced in 2018. Wells A1, B1 and B2 would be the first hydraulically fractured horizontal wells in Fields A and B. The primary driver for choosing a multi-stage completion versus traditional ‘plug and perf’ was to reduce time between stages and increase operations efficiency. Langford et al, (2015) demonstrated that this could be possible (provided there was no screen-out event) with a ball-actuated multi-stage completion, setting the deepest sleeve hydraulically, and thereafter opening frac sleeves of specific ID's with a corresponding ball approximately 0.125 inches larger than the sleeve ID. When the ball lands at the sleeve, the resulting pressure differential shears the pins in the sleeve, and shifts the sleeve to an open position, while maintaining isolation from the previous sleeve, allowing for continuous injection into a single zone. This completion type was selected for all three wells, with the following features consistent across each well:

- Total Number of Sleeves: 7 – 5 planned plus 2 contingency sleeves.
- Ball Seat Sizes: 2.625 – 3.375 inch, in 0.125 inch increments.
- Ball Sizes (not applicable to the deepest sleeve which opens hydraulically): 2.875 – 3.500 inch in 0.125 inch increments.

Well A1 in Field A would be a cemented completion, due to the expected damaged/enlarged borehole. Wells B1 and B2 would be open hole completions to keep fracture initiation pressure low, with 8 open hole packers each for zonal isolation, unless the pre-completion caliper run suggested that the hole was consistently out of gauge, in which case the cemented option would be the contingency.

Fracture Sleeve and Packer Placement

The depths of each sleeve for Well A1 (Field A) were tentatively selected prior to the operations, given that an open hole log for the existing trajectory was available. Sleeve configuration was determined primarily on reservoir quality, but also keeping in mind that the multi-stage completion might not reach TD, and therefore the final sleeve spacing still landed sleeves in good reservoir quality for several TD scenarios, even if some sleeves would not be set in the target.

Potential depths of frac sleeves for Wells B1 and B2 (Field B) would be continually reviewed during the drilling of the horizontal sections as data was available, based predominantly on reservoir quality. After the wells were drilled, and a caliper run was carried out, open hole packer depths would be selected based on hole gauge, and the final frac sleeve depths selected to be in good reservoir quality between two packers.

Hydraulic Fracturing Design Overview

Well A1 (Field A). Fracturing objectives for Well A were to connect the best quality Lower Leman sandstone units, separated by the slightly more shaley units, maximising fracture half-length, with minimum conductivity of 1000 mD.ft, and a goal of maximising near wellbore conductivity for production longevity. Detailed geomechanics models were developed for Well A1, using the two nearby offset wells that were previously hydraulically fractured, one located closer to the toe of the well, and the other closer to the heel of the well. The resulting stress profiles and estimated fluid leak-off coefficients were semi-calibrated with available injection data from existing wells prior to having onsite data, with some sensitivities considered for fluid leak-off coefficients for fluid volume planning purposes. Proppant volumes were based on what was previously, successfully placed at the offset well closer to the toe, which was 400,000 lbs of proppant, and the preliminary pump schedules included inflated pad volumes, as contingency, due to the previous high screen-out rate observed in Southern North Sea jobs. Given that the well was already drilled, further sensitivities were carried out on initiation depth within the Lower Leman sandstone, to assess fracture height growth when initiated at various locations along the existing wellbore. Preliminary designs simulated at a pump rate of 35 bpm for fractures initiating at the toe and the heel are shown in Figure 1, where the proppant coverage is shown in lb/ft^2.

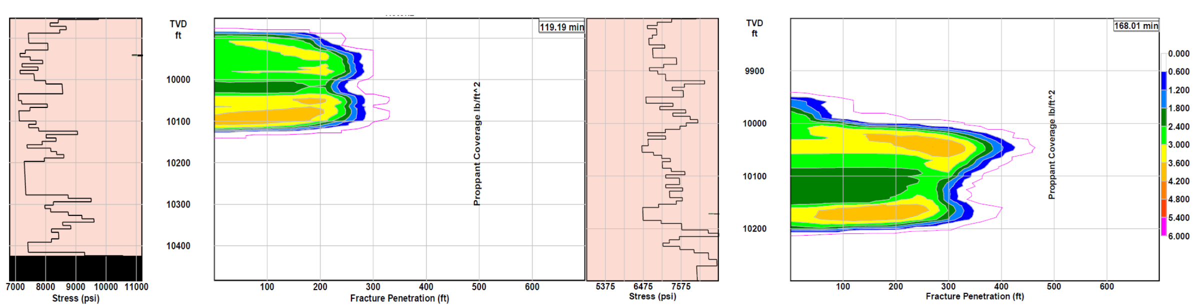

Figure 1—Preliminary resulting fracture profile towards the heel of the well (left), and towards the toe of well (right), using geomechanics developed from respective offset well data and fracture initiation depth based on existing well placement within the structure.

Wells B1 and B2 (Field B). Fracturing objectives for Wells B1 and B2 were to cover as much of the Lower Leman sandstone unit as possible, though the upper and lower most intervals tend to have a higher volume of shale. Reservoir simulations suggested that conductivity was less critical for production, however, a threshold conductivity of 750 mD.ft was agreed upon, due to expected convergent flow in the near wellbore region. Proppant volumes were based on the volume required to achieve this conductivity, while considering a conservative Pad volume, which was around 350,000 lbs of proppant. Detailed geomechanics profiles were developed for both Well B1 and B2, using the existing wells from which the horizontal wells would be sidetracked, and a third well located to the North of well B1. Resulting stress profiles and estimated fluid leak-off coefficients were semi-calibrated with available injection data from existing wells prior to having onsite data, with some sensitivities considered for fluid leak-off coefficients for fluid volume planning purposes. Fracture initiation was expected to be reasonably consistent within the best reservoir unit. Preliminary designs simulated at a pump rate of 35 bpm are shown in Figure 2, where the proppant coverage is shown in lb/ft^2, for both Well B1 and B2.

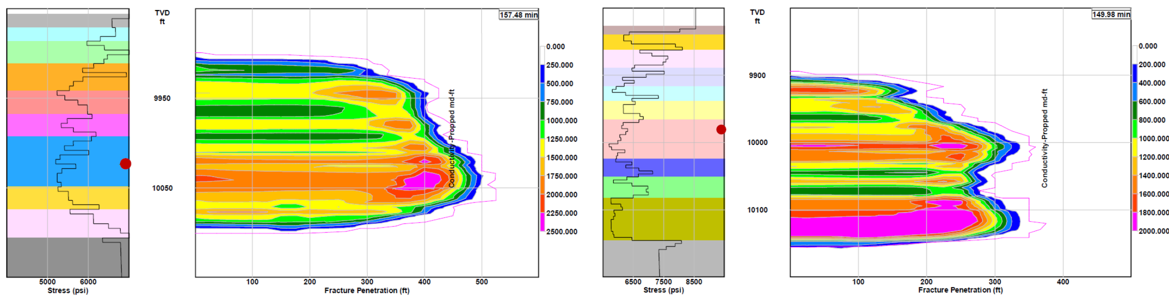

Figure 2—Preliminary resulting fracture profiles in Well B1 (left), and B2 (right), using geomechanics developed from respective offset wells.

Fracturing Materials

Fracturing Fluid. Reservoir temperatures for both Fields A and B are 188 and 213°F respectively. Historic treatments in Fields A and B used 40 – 50 lb Borate or Zirconate crosslinked fluids, with a seawater base fluid. More recently, typical fluids used in the Southern North Sea region tend to be 30-40 lb borate crosslinked fluids with a fresh water and 2% KCl base fluid, to limit the potential for scaling issues. The fluid developed used a similar approach, and the fracturing fluid system for Field A was a 40 lb gel. Optimisation of the gel loading led to a 35 lb borate crosslinked gel fluid baseline in time for the subsequent campaigns in Field B.

The freshwater replenishment strategy for each 5 stage hydraulic fracturing campaign involved bringing two fully loaded supply vessels from the closest harbour, Yarmouth, and would be transferred to the rig tanks, for bunkering to the stimulation vessel during the campaigns when required. However, during the campaign for Field A, contamination of several water holding tanks meant that the water quality was not dependable, and the strategy was revised to make water using a Reverse Osmosis unit onboard the stimulation vessel, which reliably produced good quality water for the fracturing fluid.

Proppant. The proppant to be used across all three wells was a 16/20 mesh size, fully resin coated, light weight proppant (LWP), appropriate for the maximum effective stresses that might be seen in all wells. The proppant ordering strategy was to be phased, to avoid having a large volume of proppant remaining if not all of the 15 planned stages (up to a maximum of 21 stages across all three wells) were executed.

Planned Operations

Each campaign would be executed on-line with the assigned rig on location. The stimulation vessel would connect to the rig via a single high pressure hose, positioned on a hose hanger, with a maximum pump rate of 37 bpm, sufficient for the designed rate of 35 bpm. Fracturing treating iron was rigged up from the hose hanger to a ball dropper on top of the wellhead. Both coiled tubing and well test packages were mobilised in case of screen-out, and to start well clean-up operations after the campaigns. Well A1 in Field A was to be executed first, in June 2023, in a single sailing, with the stimulation vessel to be loaded with 2.4 million lbs proppant and all chemicals required onboard for the well. Wells B1 and B2 were to be executed back-to-back in October 2023. A total of 1.75 million lbs of proppant was planned for each of Wells B1 and B2, and one return to port planned to re-load the vessel with proppant and chemicals while the rig switched treating iron, pressure control equipment and the ball dropper assembly between the wells. However, the strategy for Wells B1 and B2 was revised, due to the final completion of B2 differing from the original design (discussed later in the paper).

Ball Dropping Considerations

Recent experience at the time of planning, and lessons learned in similar North Sea operations detailed by Channa et al, (2023), contributed significantly to highlighting risks of the ball landing in proppant, or over flushing the proppant if the ball was launched late. An amended ball dropping procedure was implemented, in which under no circumstances would a ball be dropped during a main hydraulic proppant fracture treatment, to completely remove the risk of these consequences. In every scenario, the ball would be dropped to open the subsequent stage, after the downhole pressure had declined to below the estimated fracture closure pressure. The zero barrel under flush strategy was also brought forward to these operations.

The balls in the selected completion are dissolvable with exposure to temperature and salinity. Though dissolvable balls bring an unquestionable improvement in multistage fracturing, providing streamlined operations, their capacity to seal on the seat depends on its changing diameter and integrity. The dissolvable balls start to degrade once launched and immersed in the completion fluid. The rate of degradation is based on the completion fluid salinity and immersion fluid temperature. A continuous projection of the ball diameter and thus an accurate assessment of ball dissolution time enables better operations planning.

A ball dissolution predictor tool as shown in Figure 3 was therefore used to project the deterioration of the ball, using the live temperature data from the downhole gauge in each well. The simulator estimated the time at which the ball was expected to have diminished beyond its zonal isolation integrity limit, for example, when heating up during decline periods. If the simulator suggested that the ball might reach this threshold diameter during an injection, a second or even a third ball would be dropped for the same stage prior to that injection, to maintain isolation from the sleeves deeper in the well.

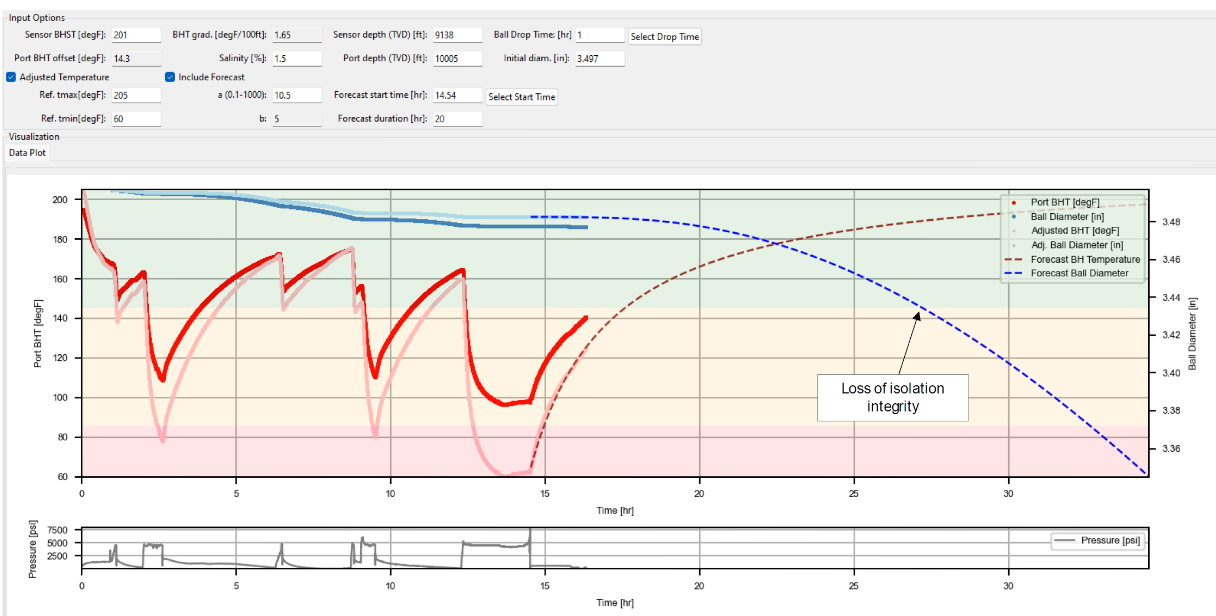

Figure 3—Ball dissolution predictor example plot, showing the forecast of the diameter with exposure to temperature.

Execution

Execution of Well A1 (Field A)

The re-completion for Well A1 started by pulling the existing tubing, and cleaning out the wellbore, including the open hole section from approximately 11,200 ft MD to 14,300 ft MD. The cleaning BHA repeatedly encountered issues passing approximately 11,900 ft MD, and a change in BHA to include a PDC reaming tool eventually allowed to pass this HUD. A further restriction was met 13,300 ft MD, and it was becoming evident that running the frac liner with the frac sleeves spaced out as per the proposed depths to

TD was unlikely. The operator and stimulation team displayed a commendable reluctance to abandoning the fracturing operations in this well a second time, and therefore, a rapid re-assessment of frac sleeve spacing was required, and would have to satisfy the two possible "what-if" scenarios, while still being positioned in reservoir of good quality in both scenarios:

1. Scenario 1: the frac liner is run to the deeper HUD at 13,300 ft MD.
2. Scenario 2: the frac liner does not reach pass the shallower HUD at 11,900 ft, in which case, the same sleeve configuration would still require the sleeves to be opposite good quality reservoir above the shallower HUD. It was accepted that in this scenario, a limited number of sleeves would be in the reservoir.

All sleeve configuration scenarios are displayed in Figure 4. Ultimately, the frac liner did not pass the shallowest HUD, and there were three frac sleeves positioned at the heel of the well, where fractures would now initiate at the roof of the Lower Leman sandstone.

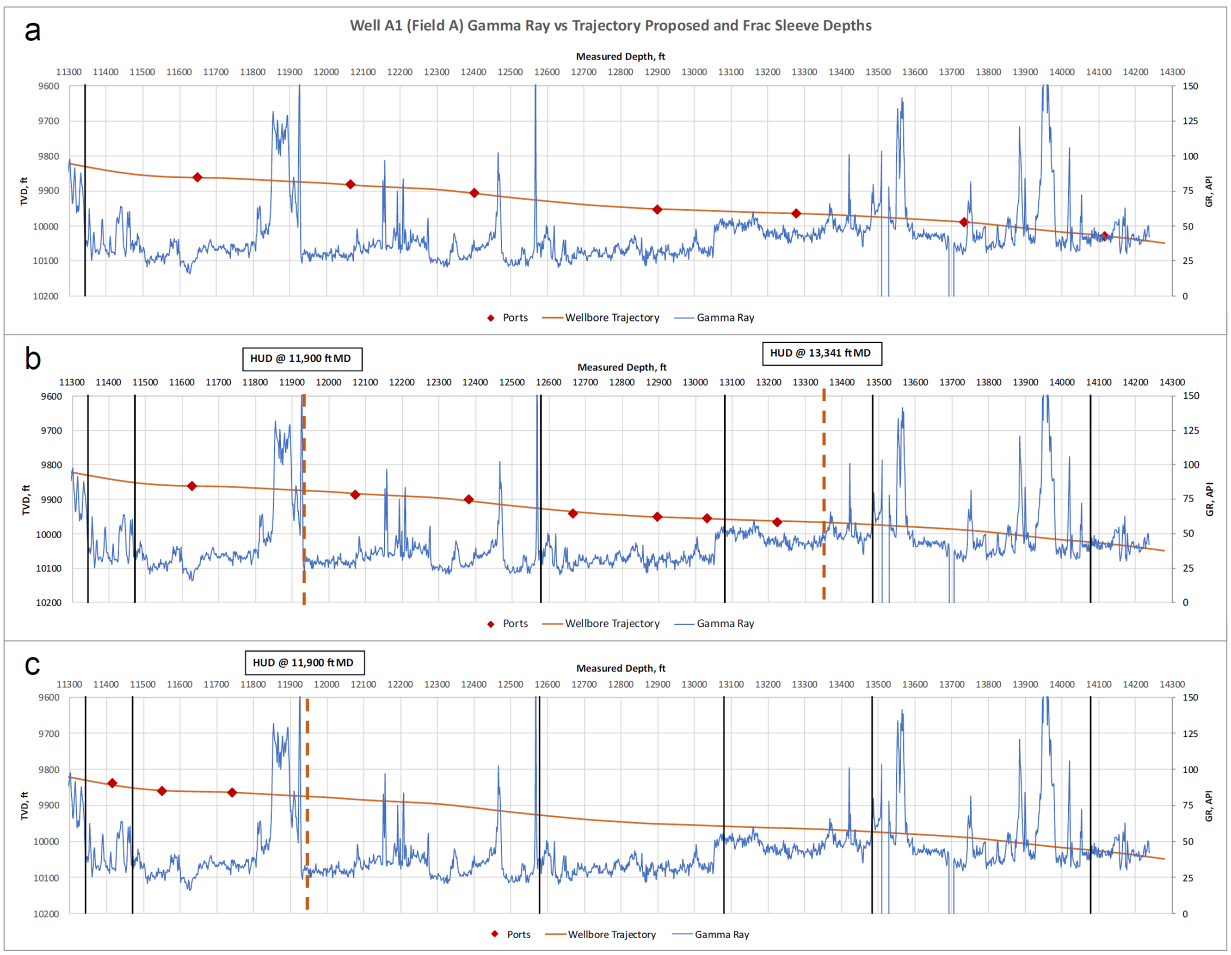

Figure 4—(a) Frac sleeve depths proposed, (b) frac sleeve depths in case the deepest HUD was reached, and (c) final frac sleeve depths where completion was set, with three sleeves in the reservoir.

During the hydraulic fracturing operations, the first and deepest frac sleeve, actuated on absolute pressure, did not show any indication of the sleeve opening. Pressure signatures of the injection tests showed the same, and a perforating run was carried out to access the reservoir. Pumping pressure trends thereafter showed signs of pushing debris, and the decline suggested poor hydraulic communication with the reservoir. Subsequently, two balls were dropped to actuate the second frac sleeve. Again, no clear signal was observed.

A decision to RIH with a CT tool for opening the seat was taken. Injections with the stimulation vessel that followed showed similar decline trends to the first sleeve. Due to time constraints, a ball was dropped for the third and final sleeve. Injections here displayed high near wellbore friction, likely due to entry friction and tortuosity. A series of sand slug injections were carried out to establish rate, reduce the friction and confirm good connection to the reservoir shown in Figure 5. Given that this was the final sleeve and last chance to fracture the reservoir, the planned proppant volume was increased to 500,000 lbs of 16/20 resin coated LWP, using the maximum fluid capacity of the stimulation vessel. Maximising near wellbore conductivity was defined as a key objective early on, and now even more critical due to the connection between the reservoir and wellbore being at the roof of the reservoir, therefore, even if it was anticipated that the full proppant volume could not be placed, a proppant ramp of 7, 8 and 9 ppa would be performed before the flush stage.

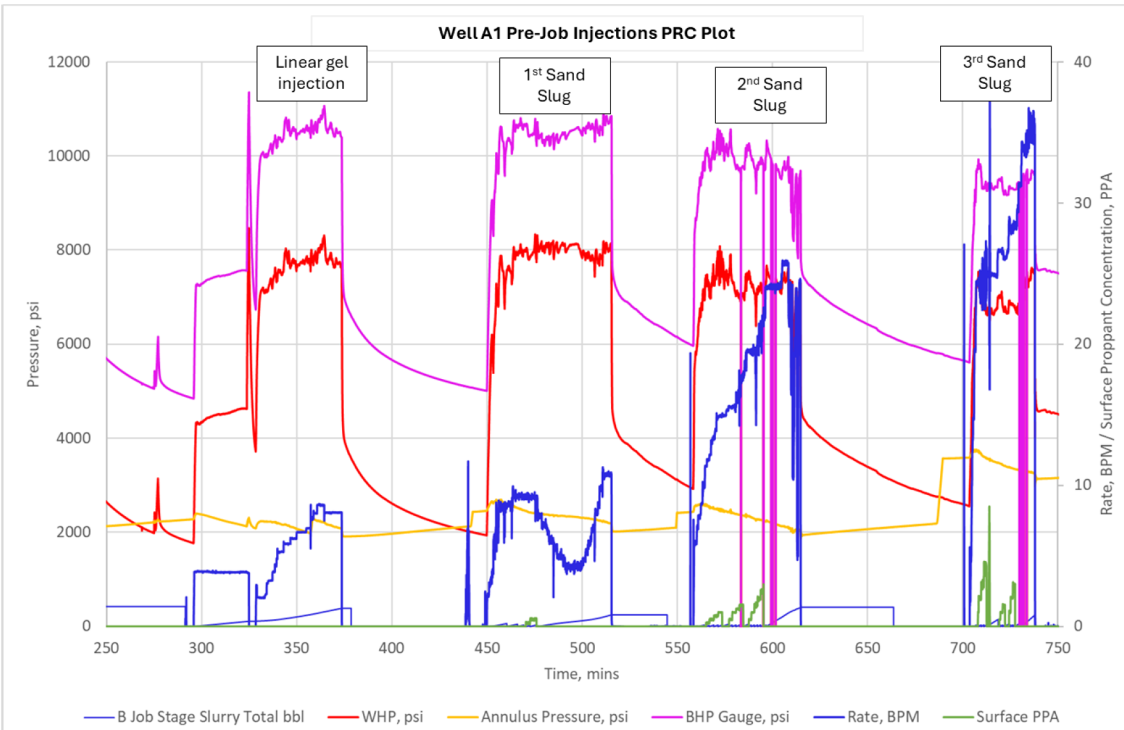

Figure 5—Well A1 multiple pre-job injections to establish rate and confirm connection with the reservoir prior to pumping the main treatment.

Pressure fluctuations were observed through the proppant stages of the main treatment (Figure 6). Pressures on surface and downhole gauges started to increase as 5 ppa reached the formation, likely due to packing the fracture. At 6 ppa, calculations indicated the pressure would increase above maximum allowable surface pressure during the flush stage. Therefore, flush was called early, being sure to still execute the ramp of high proppant concentration, leaving a slug of 9 ppa in front of the sleeve. Flush was executed with one rate drop to 28 bpm to remain below maximum allowable surface pressure. Total proppant pumped was 250,000 lbs, half of the planned volume.

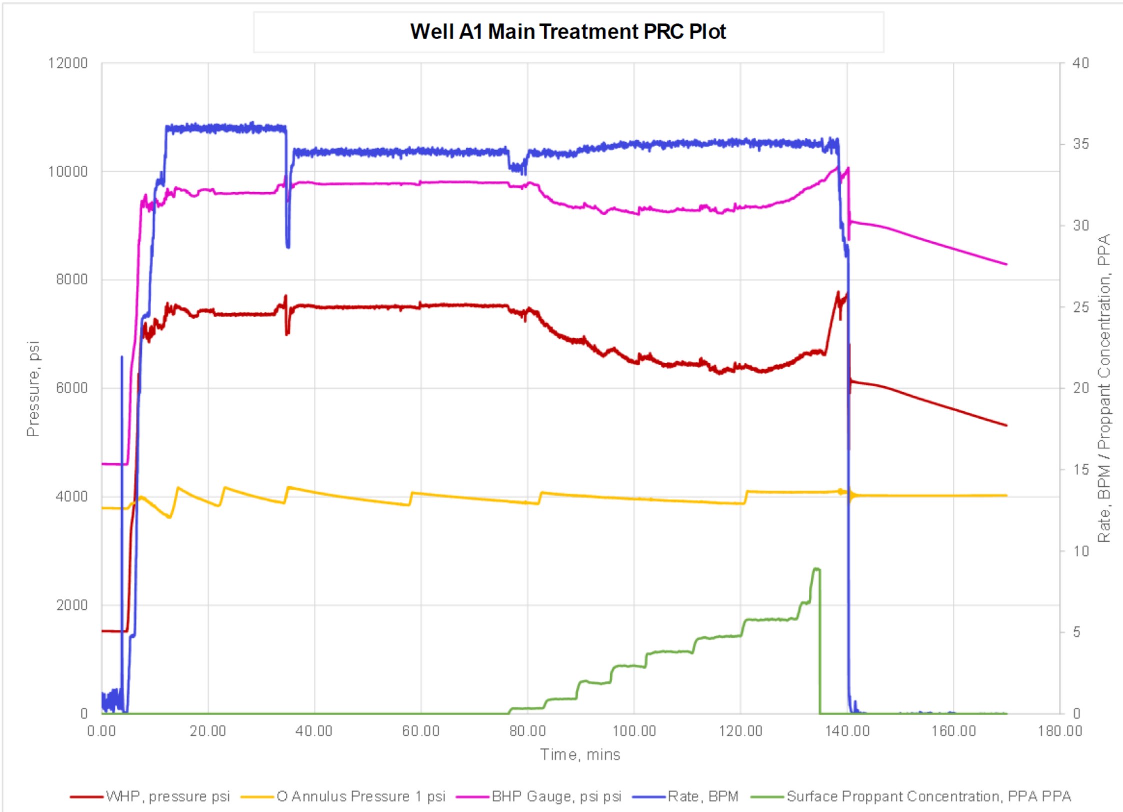

Figure 6—Well A1 main treatment plot showing increasing pressure trends in final 20/40 stages, and final high proppant concentration ramp prior to shutdown.

Execution of Well B1 (Field B)

The sidetrack of Well B1 was drilled to TD as planned. Throughout the drilling, numerous frac sleeve locations were identified based on reservoir quality. After the post-drilling caliper run, packer depths were selected based on hole gauge, with locations for all 8 open hole packers selected, and frac sleeve locations were approximately, but not strictly, central to the packer depths. The completion was run to TD without issue, with the final frac sleeve (shown in Figure 7).

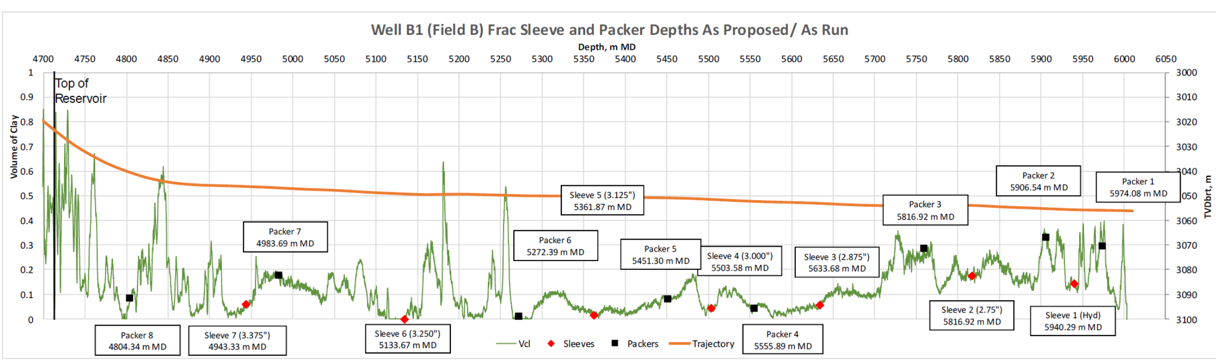

Figure 7—Well B1 (Field B) frac sleeve and open hole packer depths, run as per the original proposed depths.

Execution of the hydraulic fracturing campaign in Well B1 occurred over several weeks, due to a combination of factors, including weather conditions, as well as other ongoing fracturing operations (including Well B2). The remaining 16/20 resin coated LWP inventory from Well A1 (Field A) was utilised. During the first injections through the first port, high near wellbore friction was observed and 100 mesh sand slugs at different sand concentrations were pumped for remediation. However, high friction pressures were still observed, and a proppant admittance test (1 to 3 ppa) was executed. As the tail of the 3 ppa stage was passing through the sleeve, the treatment pressured out. This was the first indication of a misaligned of the wellbore and fracture plane. Following a second screen-out at the second frac sleeve, this time as 7 ppa hit the formation, the gel loading was increased to 40 lb, for a more robust fluid, and the maximum proppant concentration limited to 6 ppa, to avoid screen-out due to possible near wellbore restrictions. Thereafter, main fracture treatments at frac sleeves 3-6 were executed without issue, building net pressures of around 1500 – 2000 psi.

The ball dissolution tool was used at every stage to predict the requirement for another ball to maintain isolation from the previous stage. An increase in friction could be correlated at the point where the trajectory became horizontal, and thus could be timed and expected, and not mistaken for a screen-out occurrence. At the final frac sleeve, an attempt to place higher than 6 ppa would be executed, given the next operation would be a CT intervention. At the same time, a second ball was not dropped to maintain isolation on the previous stage in attempt to avoid issues during the final CT clean out run. However, the ball dissolution predictor suggested the ball diameter could degrade below the threshold and therefore the final main treatment was started without delay, to keep downhole temperature cool, and slow dissolution of the ball. Nearing the end of the main treatment, though, a sudden, sharp drop in pressure was observed, followed by a screen-out during the 6 ppa stage, suggesting that the ball had indeed failed, and passed through to the next seat causing the initial pressure drop, and the swift screen-out response thereafter. A total of approximately 1.9 million lbs of proppant was placed in Well B1 (Field B), with minimal remaining proppant inventory. Major events and proppant volumes for injections carried out in all seven frac sleeves are summarised in Table 3.

Table 3—Well B1 (Field B) Hydraulic Fracture Execution Summary

Frac Sleeve	Job Rate, BPM	16/20 RC-LWP Volume, k lbs	Max. Proppant Concentration, ppa	Screen-out Event	No. Balls Dropped	Comments
1	35	7	3	Yes	N/A	Pressure out in proppant slug, likely due to high tortuosity / misalignment of fracture and wellbore. CT clean out required.
2	35	251	7	Yes	2	Screen-out as 7 ppa hit formation. CT required.
3	35	345	6	No	2	Increased gel loading to 40 lb, and max 6 ppa. Executed as planned.
4	35	374	6	No	2	40 lb gel / 6 ppa. Executed as planned. Rapid bleed down in annulus pressure caused proppant to be drawn out of fracture, and set plug. CT clean up required.
5	35	385	6	No	3	40 lb gel / 6 ppa. Executed as planned.
6	35	357	6	No	2	40 lb gel / 6 ppa. Executed as planned.
7	35	238	6 *(Planned 7-8-9 ppa ramp, not executed)*	Yes	1	40 lb gel. Re-designed treatment to attempt 7-8-9 ppa, though not executed due to screen-out. Ball failure at seat, no second ball dropped (to avoid issues with CT during the imminent clean out).

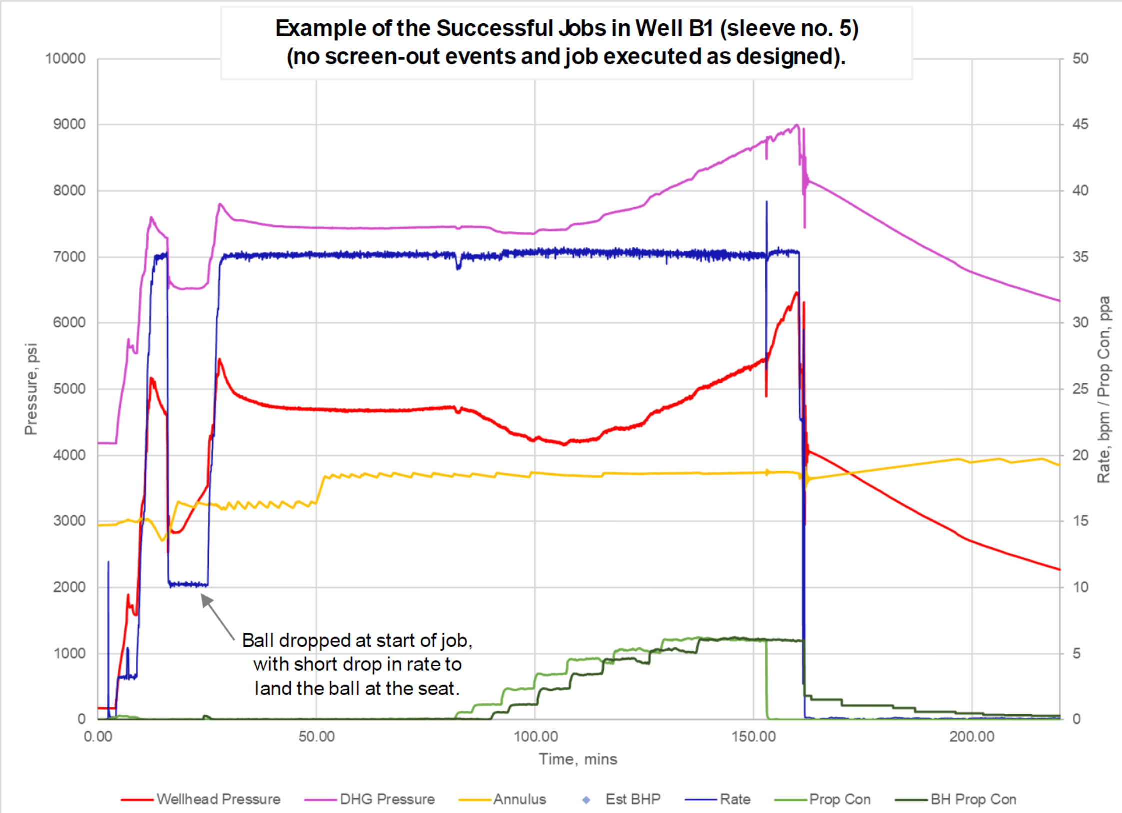

Figure 8—Well B1 main treatment executed without screen-out event for sleeve 5. Similar treatment schedules executed at sleeves 3, 4 and 6.

Execution of Well B2 (Field B)

Execution of the hydraulic fracturing campaign in Well B2 occurred in two phases; two injection tests carried out between frac sleeve 5 and 6 in Well B1 prior to a sail to port to wait out bad weather and reload with some additional proppant and chemicals, and another injection test followed by a main treatment carried out after Well B1 had been completed.

The final completion for Well B2 was not as designed. The well was drilled to the planned TD, though unfortunately, the drill pipe became stuck. Several attempts were made to recover the pipe, but to no avail. Eventually, a chemical cut was made at a depth that met the criteria for sleeve placement, that is, good reservoir quality in the main target unit and low volume of clay. A final attempt was made to pull the drill pipe above the cut, though was never recovered.

A flow trial was carried out for Well B2 that showed promising rates. However, due to the tight nature of the reservoir, the rates were expected to rapidly decline. In order to avoid the same circumstances that occurred with Well A1 (Field A), that is, having to come back at a later date to hydraulically fracture, the decision was made to attempt to place at least a single hydraulic fracture in Well B2 via the chemical cut and through approximately 5,900 ft of drill pipe, now the lower completion. The fishing tool was modified onboard to incorporate an elongated seal and a crossover, to make up the premium connection between tubing and drill pipe fish, providing hydraulic and mechanical continuity between the upper and lower completions. The smaller ID of the drill pipe and the numerous tool joint ID's with an even smaller diameter would mean higher surface pressures due to friction were expected.

The initial phase of Well B2 operations included several injection tests, introducing different fluids into the completion using a step-by-step approach: a breakdown with linear fluid, followed by a crosslinked gel pill to observe pressure response at formation, followed by a conservative, low concentration sand slug with 100 mesh sand. Pressure fluctuations were observed as the sand hit the formation, and an unusual pressure trend observed after shutdown (Figure 9). Planned proppant size 16/20 resin coated LWP was deemed an unnecessary risk. Therefore, during the sail to re-load proppant after these injection tests, a volume of 283,000 lbs 20/40 resin coated LWP stock was quickly sourced and loaded.

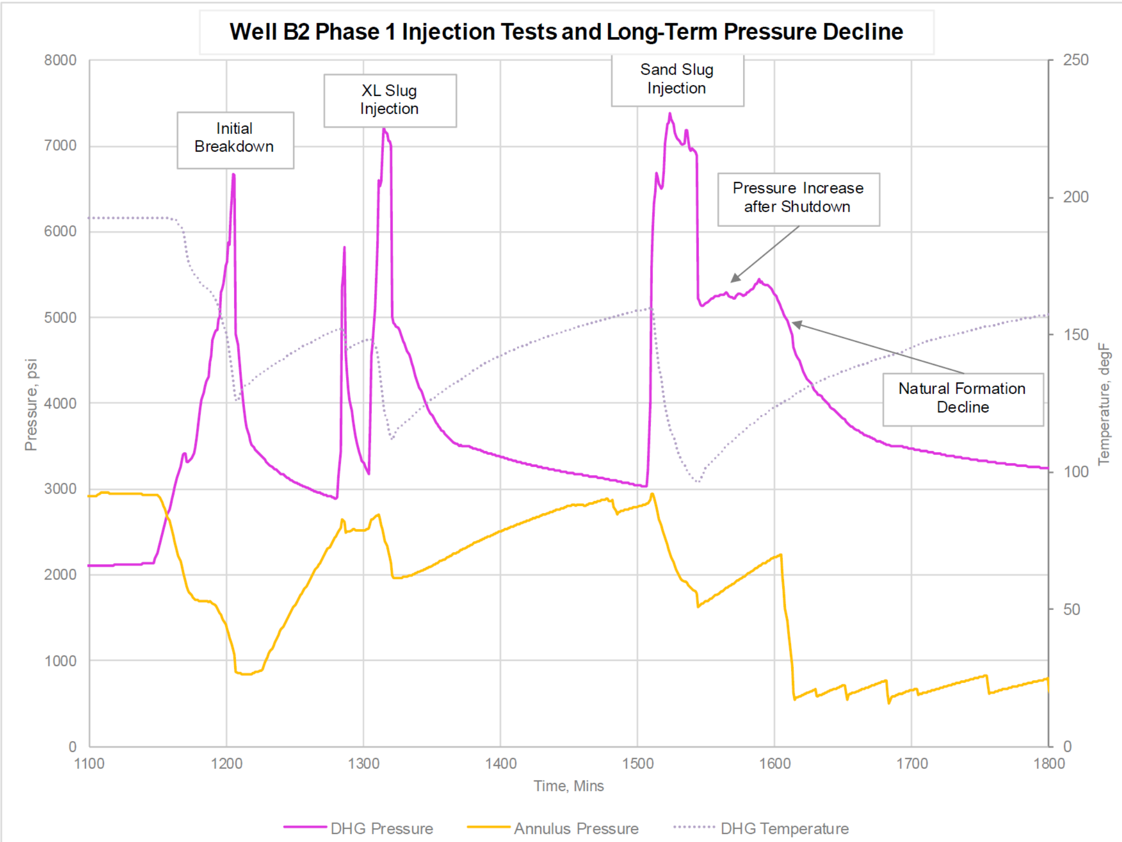

Figure 9—Well B2 pressure response during first execution phase, showing packing trend after shutdown of the final injection, and eventually the natural formation pressure decline.

The second phase of Well B2 operations repeated an injection test to re-confirm injectivity, inclusive of sand slugs for assurance. Treating pressures were higher than previously observed and attributed to additional friction, however, a water hammer was still observed at shutdown, a good confirmation of hydraulic communication with the reservoir. The full injection was carried out with crosslinked gel, and a decline analysis carried out to gain the fracture closure pressure and an estimate of leak-off coefficient. The job design planned to pump the entire volume of 20/40 resin coated LWP, followed by another 200,000 lbs of the available 16/20 resin coated LWP from Well B1 if pressure responses were favourable. An engineered over displacement was also planned, of 5 bbl, to avoid having to run CT in an already complex completion.

During the treatment shown in Figure 10, it was observed that similar friction responses to Well B1were observed as each new proppant concentration front reached a certain point in the completion, which was later confirmed to be the top of the drill pipe fish. Due to the smaller ID of 5,900 ft of drill pipe, and the presence of the narrower tool joints, the change in friction was much greater, and the treating pressures crept up much quicker. One pressure anomaly was observed inconsistent with the expected trends for friction, and the rate had to be dropped from 35 bpm to 28 bpm to manage the pressures, after which an increase in pressure thought to be proppant packing was observed. Transitioning to larger proppant was deemed high risk, and so only the 20/40 proppant was cleared from the vessel before displacing the full wellbore volume and an additional 5 bbl to remove proppant from the completion. A final steep drop in pressure was observed after the proppant was cleared from the completion, due to loss of friction as the last proppant was displaced from the completion. A total of 283,00 lbs of 20/40 resin coated LWP proppant was placed.

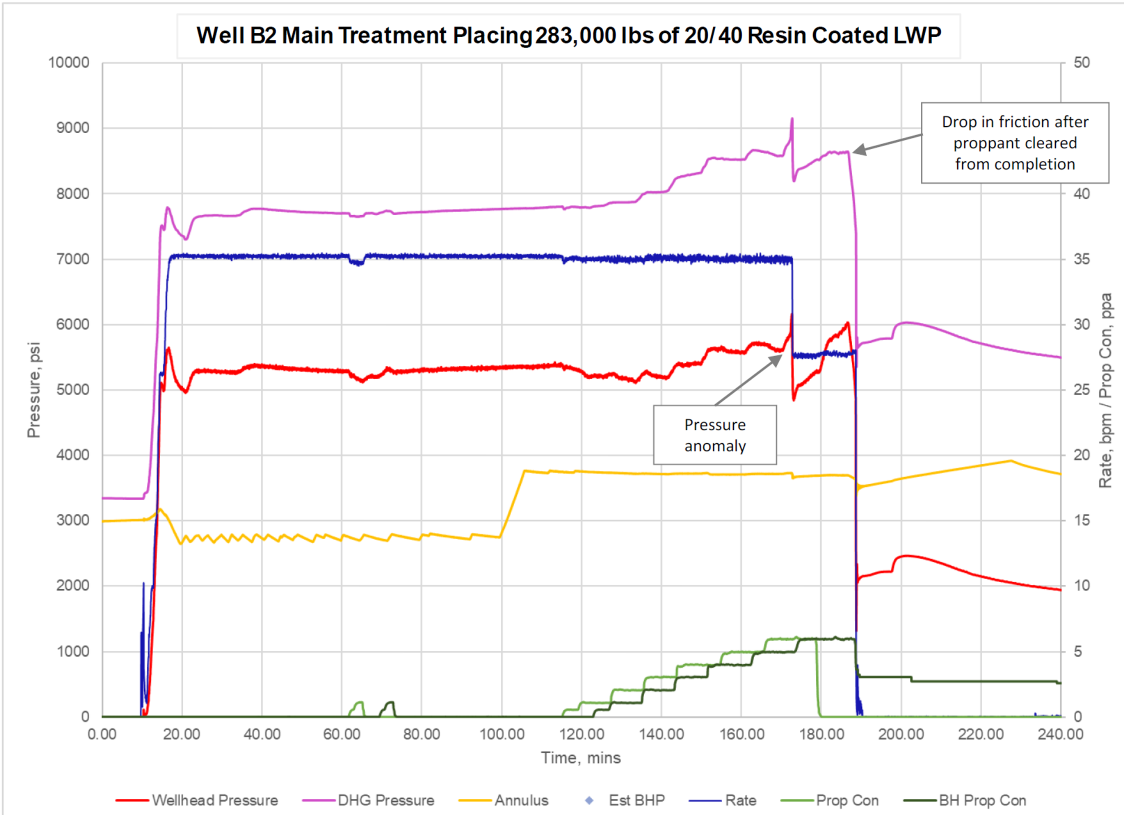

Figure 10—Well B2 main treatment pumping 283,000 lbs 20/40 proppant into formation via the severed drill pipe.

Results

Results of Well A1 (Field A)

Only a single fracture treatment was executed in Well A1, out of a planned 5. Post-treatment evaluation suggested that the treatment may have covered the upper two units of the best reservoir quality targets, and covering approximately 50% of the Lower Leman sandstone. The production uplift post-stimulation met the P90 forecasted rates, as shown in Figure 11, with a single hydraulic fracture at the heel of the well. Given the difficulties running the completion and injecting into the deeper frac sleeves that were set in the reservoir, this was an excellent result for the well, proving that hydraulic fracturing will contribute to maximising production in this part of the field, and providing a basis for future field development through hydraulic fracturing.

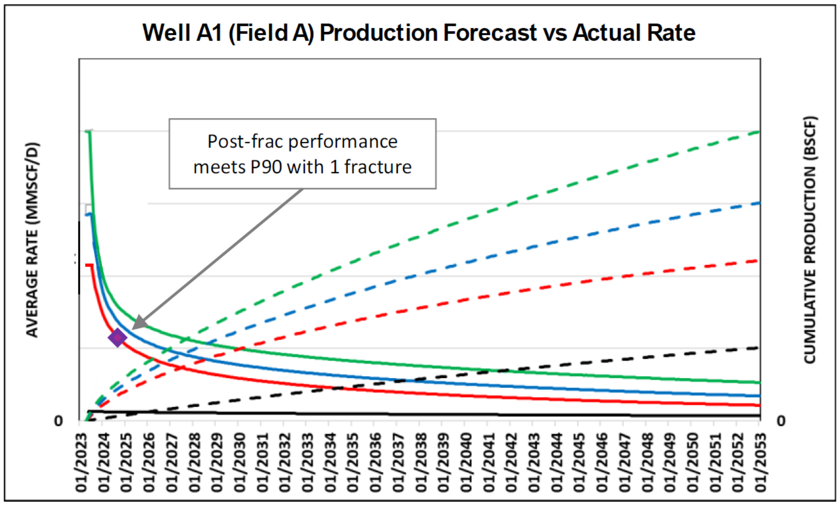

Figure 11—Well A1 (Field A) post-stimulation production, with production rate in line with the P90 estimate with just one hydraulic fracture.

Results of Well B1 (Field B)

Well B1 in Field B has injections at every frac sleeve. Post clean-out, the well came online far exceeding the forecasted P10 rates, shown in Figure 12. After one year on production, rates have declined as expected, though still meet the P50 forecasted rates.

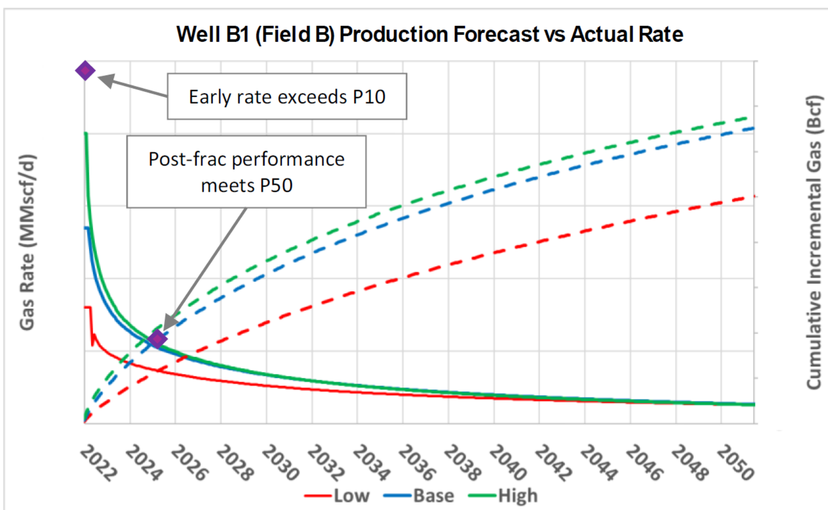

Figure 12—Well B1 (Field B) post stimulation production, with early rates shown as exceeding forecasted rates, and later declining to match the P50 expected rates.

Results of Well B2 (Field B)

Well B2 in Field B had no CT intervention and instead had an engineered over displacement to mitigate proppant flowback. The resulting production was perhaps the most impressive of all three wells, showing rates exceeding the P90, with a single hydraulic fracture, less proppant volume overall placed, smaller proppant mesh size, and an over displacement (Figure 13).

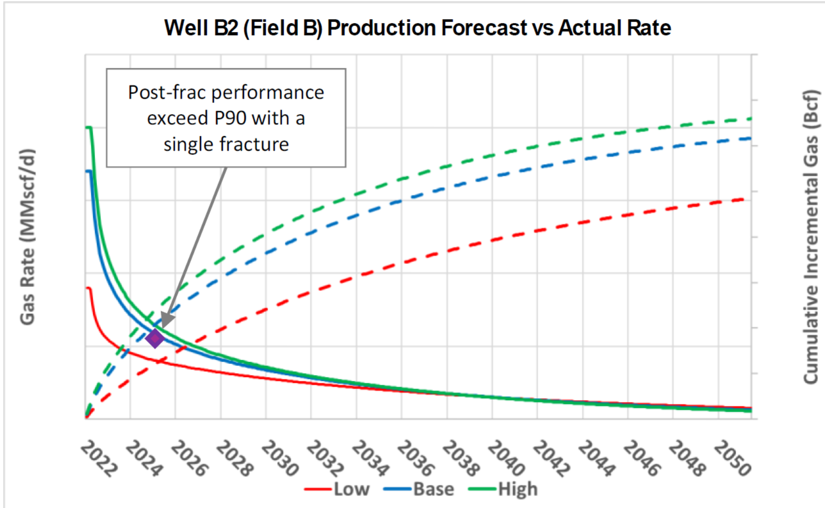

Figure 13—Well B2 (Field B) post stimulation production, with rate exceeding P90 forecast, close to P50 rates, with a single hydraulic fracture.

Lessons Learned

A total of 8 main fracture treatments were executed out of a planned minimum of 15 treatments. Nevertheless, production results were positive in all three wells. Decisions made throughout the planning phase, as well as those real-time decisions made to address operational challenges faced, all contributed to the successful placement of hydraulic fractures in each well, especially those wells where only one fracture could be executed. Key learnings can be summarised as:

- Although potentially time-consuming, considering various "what-if" scenarios during the planning can significantly speed up decision-making during online operations.
- Full understanding of the target reservoirs and different fracture initiation depths vertically in the pay through detailed analyses of offset wells is beneficial in managing real time decisions and optimising the placement of the frac sleeves when the as-planned tally cannot be run to TD.
- Careful resource planning over several campaigns can reduce overall campaign cost and minimise remaining proppant and chemical inventories.
- Implementation of ball drop learnings from previous campaigns were key to removing unnecessary risks associated with dropping a ball during the flush of a main treatment, reducing CT intervention requirements. The ball dissolution predictor proved reliable, and critical to maintaining isolation integrity from previous stages.
- Reluctance to abandon operations due to challenges in Well A1 (Field A), and perseverance to deliver a low case completion scenario led to the successful placement of one hydraulic fracture with excellent production results, and provided critical input for future field development strategies.
- The final and vastly innovative completion solution put forward for Well B2 (Field B) was a clear example of the operator learning from their past experiences, and electing to do their utmost to execute at least one hydraulic fracture treatment while having a stimulation vessel on hire, rather than abandoning the operations, as was originally done with Well A1. The results showed that this was decision indeed was worthwhile.
- While not typically executed in the North Sea region, the engineered over displacement for Well B2 (Field B) removed requirement for post-fracture CT intervention in a complex completion. The impact on production is thought to be minimal, due to the very small over displacement volume, and positive post-fracture production result.

Future Considerations

The selection of a ball-actuated completion removed requirement for several CT interventions for at least one well, though in a screen-out event, does still require a CT operation. Therefore, a screen-out, while not necessarily negative for the fracture and production, will always reduce the operations efficiency in a horizontal well with the same completion. Newest multi-stage hydraulic fracturing execution methods that make use of a drill pipe to manipulate frac sleeves have proven to reduce time in all fracturing scenarios, with a much faster screen-out recovery by reverse circulation of proppant laden slurry through the drill string. These completions may be considered in future multi-stage horizontal wells in Fields A and B to further improve operational efficiency in the offshore North Sea high stakes environment.

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Abbreviations

BHA	Bottom Hole Assembly
bpm	Barrels Per Minute (relating to fracture treatment pump rate)
CT	Coiled Tubing
HUD	Hold Up Depth
ID	Internal Diameter (relating to tubulars)
KCl	Potassium Chloride (relating to brine)
lbs	Pounds (relating to proppant mass)
LWD	Logging While Drilling
LWP	Light Weight Proppant
RIH	Run In Hole
ppa	Pounds (lb) of Proppant Added (relating to proppant concentration)
TD	Total Depth

References

Bocaneala, B., Barrett, C., Holland, B., et al 2015. The Evolution of Completion Practices and Reservoir Stimulation Techniques in the North Sea. Presented at the SPE Offshore Europe Oil and Gas Conference and Exhibition, Aberdeen, UK, 8-11 September. SPE-175443-MS. https://doi.org/10.2118/175443-MS.

Channa, S., Bocaneala, B., Wood, P., Tolson, K., and Heidenreich, B. Successful Execution of Hydraulic Fracturing via a Multi-Stage Ball Drop Completion to Extend Platform Life and Access Previously Untouched Reserves in a Challenging Operational Environment – A Central North Sea Case Study. Presented at SPE International Hydraulic Fracturing Technology Conference and Exhibition, Muscat, Sultanate of Oman, September 2023. SPE-215641-MS. https://doi.org/10.2118/215641-MS.

Langford, M.E., Holland, B., Green, C.A. 2013. Offshore Horizontal Well Fracturing: Operational Optimisation in the Southern North Sea. Presented at SPE Offshore Europe Oil and Gas Conference and Exhibition, Aberdeen, UK, 3-6 September. SPE-166550-MS. http://doi.org/10.2118/166550-MS.